

Mathematical Equations Used in NER-Augmented Machine Translation Experiment

1. BLEU Score

Bilingual Evaluation Understudy — measures n-gram overlap between hypothesis and reference translations

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

where p_n = modified n-gram precision, $w_n = \frac{1}{N}$ (uniform weights), $N=4$

2. Brevity Penalty (BP)

Penalizes translations shorter than the reference.

$$\text{BP} = 1 \quad \text{if } c > r ; \quad e^{(1-r/c)} \quad \text{if } c \leq r$$

where c = hypothesis length, r = reference length

3. Modified N-gram Precision

Clipped count of matching n-grams to avoid overcounting

$$p_n = \frac{\sum_C \sum_{\text{ngram} \in C} \text{Count}_{\text{clip}}(\text{ngram})}{\sum_C \sum_{\text{ngram} \in C} \text{Count}(\text{ngram})}$$

$\text{Count}_{\text{clip}} = \min(\text{Count}(\text{ngram}), \text{MaxRefCount}(\text{ngram}))$

4. chrF Score

Character-level F-score using character n-grams.

$$\text{chrF}_\beta = (1 + \beta^2) \cdot \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}}$$

where chrP = char n-gram precision, chrR = char n-gram recall, $\beta=2$

5. NER Tag Injection Function

Augments source sentence by wrapping recognized entities with type tags.

$$f_{\text{NER}}(s, E) = s' : e_i \rightarrow \langle T_i \rangle e_i \langle /T_i \rangle \quad \forall (e_i, T_i) \in E$$

$E = \{(e_1, T_1), (e_2, T_2), \dots, (e_k, T_k)\}$ = set of entities and NER labels

6. Cross-Entropy Loss (Training Objective)

Minimized during Seq2Seq model training

$$\mathcal{L} = -\frac{1}{|Y|} \sum_{t=1}^{|Y|} \log P_\theta(y_t \mid y_{<t}, X)$$

X = source tokens, $Y = (y_1, \dots, y_{|Y|})$ = target tokens, θ = model parameters

7. Scaled Dot-Product Attention (Transformer)

Core mechanism of the MarianMT encoder-decoder architecture.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

Q = queries, K = keys, V = values, d_k = key dimension

8. Multi-Head Attention

Allows the model to attend to different representation subspaces.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

9. Beam Search Decoding

Selects the translation with highest probability

$$\hat{Y} = \arg \max_{Y \in \mathcal{B}} \sum_{t=1}^{|Y|} \log P_\theta(y_t \mid y_{<t}, X)$$

$\mathcal{B}$ = set of candidate sequences (beam width = 4 in our experiment)

10. Bootstrap Resampling (Statistical Significance)

Tests whether BLEU improvement is statistically significant.

$$\delta_b^* = \text{BLEU}(H_{\text{NER}}^{*b}, R^{*b}) - \text{BLEU}(H_{\text{base}}^{*b}, R^{*b}), \quad b = 1, \dots, B$$

$p = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\delta_b^* \leq 0]$, reject H_0 if $p < \alpha$ ($B=1000, \alpha=0.05$)

11. Confidence Interval for BLEU Difference

Reports the range within which the true improvement lies.

$$\text{CI}_{95\%} = [\delta_{(0.025)}^*, \delta_{(0.975)}^*]$$

Our result: $\Delta\text{BLEU} = +0.33$, $\text{CI} = [+0.136, +0.530]$, $p=0.001$